A Comprehensive **Review** of Empirical and Dynamic **Wildfire** Simulators and Machine Learning Techniques used for the **Prediction** of **Wildfire** in Australia

H Singh, LM Ang, D Paudyal, M Acuna… - Technology, Knowledge …, 2025 - Springer

… better anticipate and mitigate the impacts of **wildfire spread**. This **review** will discuss the range of empirical and dynamic tools used for **wildfire prediction**, highlighting their advantages, …

☆ Save 99 Cite Cited by 3 Related articles All 4 versions

A **review** of recent advances in risk analysis for **wildfire management**

C Miller, AA Ager - International journal of **wildland fire**, 2012 - CSIRO Publishing

… We also **review** recent advances in addressing temporal **dynamics** of fire risk and spatial optimisation of fuels **management** … for future research and development in **wildfire** risk analysis. …

☆ Save 99 Cite Cited by 347 Related articles All 11 versions

Facing the **wildfire spread** risk challenge: where are we now and where are we going?

J Sun, W Qi, Y Huang, C Xu, W Yang - Fire, 2023 - mdpi.com

… **review** and explore further research in this field. This article primarily focuses on elucidating and discussing **wildfire spread behavior** … following results for **wildfire spread simulation**. In …

☆ Save 99 Cite Cited by 25 Related articles All 6 versions »

Mathematical models and calculation systems for the study of **wildland fire** behaviour

E Pastor, L Zárate, E Planas, J Arnaldos - Progress in Energy and …, 2003 - Elsevier

This is a **review** of the most important work in **wildland fire** mathematical modelling which … **wildland fire** models to be sorted. Surface fire **spread** models, crown fire initiation and **spread** …

☆ Save 99 Cite Cited by 545 Related articles All 10 versions

A **review** of **wildland fire spread** modelling, 1990-present, 1: Physical and quasi-physical models

AL Sullivan - arXiv preprint arXiv:0706.3074, 2007 - arxiv.org

… Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial scales, IUSTI concentrates on resolving the chemical and conservation equations at a …

☆ Save 99 Cite Cited by 62 Related articles All 4 versions »

Operational **wildfire** suppression modelling: a **review** evaluating development, state of the art and future directions

TJ Duff, KG Tolhurst - International Journal of **Wildland Fire**, 2015 - CSIRO Publishing

… **review**, we focus on the development and use of models that emulate fire suppression for decision support. We **review** … There are two main approaches to dynamic fire **spread simulation**…

☆ Save 99 Cite Cited by 92 Related articles All 7 versions

A **review** of **wildland fire spread** modelling, 1990-present 2: Empirical and quasi-empirical models

AL Sullivan - arXiv preprint arXiv:0706.4128, 2007 - arxiv.org

… to **model** the **spread** and behaviour of wildland fires across the landscape. This series … **review** papers endeavours to critically and comprehensively **review** all types of surface fire **spread** …

☆ Save 99 Cite Cited by 42 Related articles All 4 versions »

Wildland surface fire **spread** modelling, 1990–2007. 3: **Simulation** and mathematical analogue models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… The present series of articles endeavours to comprehensively **survey** and précis all types of surface fire **spread** models developed during the period 1990–2007. The present paper …

☆ Save 99 Cite Cited by 462 Related articles All 5 versions

Modelling smoke transport from wildland fires: a **review**

SL Goodrick, GL Achtemeier, NK Larkin… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… Another grid **model** receiving attention for **wildland fire** related air quality issues is the … of the modelling studies cited in this **review** involves the **dynamics** of the buoyant phase of the …

☆ Save 99 Cite Cited by 149 Related articles All 8 versions

DEVS-FIRE: Towards an integrated **simulation** environment for surface **wildfire spread** and containment

L Ntaimo, X Hu, Y Sun - **Simulation**, 2008 - journals.sagepub.com

… applied to an example of fire **spread simulation** [44]. A recent … the **wildfire spread** and containment **simulation model** presented in … We **review** the basics of **wildfire behavior** [56] to set …

☆ Save 99 Cite Cited by 122 Related articles All 9 versions